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Investigation of Turbulence and Turbulent Prandtl Number Models for He-Xe Thermal Hydraulics in Quasi-Triangular Channel

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Abstract

Compact nuclear reactor systems usually use helium–xenon (He-Xe) mixtures as coolants. Tight-lattice rod-bundled channels, serving as primary core configurations in compact nuclear reactor designs, exhibit quasi-triangular cross-sections where fluid dynamics substantially deviate from circular tube behavior. This study evaluates the applicability of turbulence models and turbulent Prandtl number (Pr_t) models in quasi-triangular channels through systematic numerical simulations. The results demonstrate that the Transition SST model accurately resolves flow dynamics and turbulence development in helium–xenon mixtures, while implementing Pr_t models significantly enhances temperature prediction accuracy. Among the evaluated models, the Weigand model achieves optimal performance by dynamically adapting Pr_t values across flow regimes. Further refinements targeting parameters governing near-wall Pr_t distribution are identified as critical pathways for improving numerical simulation precision of low-Prandtl-number fluids in geometrically complex nuclear systems.

Keywords: helium–xenon gas mixture; low-Prandtl-number fluid; turbulence model; turbulent Prandtl number model; CFD

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1. Introduction

Compact reactors represent a pivotal direction for nuclear energy development, characterized by structural compactness, low mass, and high energy density. These systems address energy demands for sustained high-power operations in extreme terrestrial, deep-sea, and space environments. The Brayton cycle is the ideal thermodynamic conversion technology for this class of reactors [1], owing to its high efficiency, simple cycle configuration, and high power density. Using working fluids such as helium gas offers excellent thermodynamic properties and chemical stability, minimizing corrosion and enabling tolerance of higher inlet temperatures. Consequently, the helium Brayton cycle possesses strong application potential.

However, helium's low molar mass and poor compressibility present engineering challenges. These properties necessitate turbomachinery and heat exchangers that are large-scale and require multiple stages. When mixing He with heavier noble gases, the thermal properties of the binary gas mixture are typically superior to those of pure gases of equal molecular weight of the mixture. Blending xenon (molar concentration: 8.6–30.0%) into helium as the Brayton cycle working fluid [2], He-Xe demonstrates low aerodynamic loading, high chemical stability, superior heat transfer performance, and favorable compressibility [3,4]. These properties enable substantial reductions in system volume and mass, facilitating advanced compact integration [5,6]. For example, closed Brayton cycle (CBC) turbo-machines for He working fluid typically have between three and four times the number of stages of those designed for He-Xe [5].

Nuclear power systems employing He-Xe to directly drive closed Brayton cycle turbines constitute a good solution for compact reactors in extreme environments, exhibiting significant advantages. For example, in space applications, gas-cooled reactors utilizing He-Xe mixtures as coolants operate at elevated outlet temperatures [1], which leads to high efficiency. Second, based on the aforementioned discussion, nuclear reactors employing CBC systems that utilize He-Xe as the working fluid exhibit advantages in both compact integration and reduced system mass. Due to He-Xe's chemical inertness, corrosion problems can be eliminated, thereby extending the service life of nuclear reactors [2]. For He-Xe gas-cooled reactors, the thermal–hydraulic mechanisms of the coolant within the reactor core are critical for thermal–hydraulic safety analysis and compact design optimization. In nuclear reactor design, the ability to maximize efficiency while preventing safety incidents is contingent upon a thorough understanding of the coolant's thermal–hydraulic mechanisms. In safety analyses under reactor accident conditions, a thorough understanding of the coolant's flow and heat transfer characteristics enables more effective design of heat removal and subsequent accident management, thereby avoiding severe consequences.

The low Prandtl number (Pr~0.23) of He-Xe mixtures distinguishes their heat transfer characteristics from pressurized water (Pr $\geq$ 0.7) used in the light water reactors. To elucidate these mechanisms, Taylor et al. [7] and Qin et al. [8] conducted experimental studies on He-Xe flow in circular tubes, establishing the flow and heat transfer correlations. Further optimization of He-Xe gas-cooled reactor applications has led to diverse core structure designs, exemplified by the Prometheus program [9]. Among these, tight-lattice rod-bundled channel design has emerged as the mainstream option due to its compactness and low mass [10]. The quasi-triangular channel is an extreme approximation of the tight-lattice rod bundle. The flow and heat transfer characteristics of He-Xe in this channel more closely resemble those in the reactor core. Compared to circular tubes, the flow structure in quasi-triangular channels is significantly more complex, with flow maldistribution and stagnant regions in narrow gaps. In circular tubes, viscous-dominated regions are confined to thin boundary layers near walls, whereas in quasi-triangular channels, boundary layer interactions in narrow gaps induce flow stagnation, where laminar flow persists even under high Reynolds numbers (Re), while the core region transitions to turbulence [11]. Consequently, He-Xe's flow and heat transfer conclusions derived from circular tube studies may not apply to quasi-triangular geometries.

Several experimental studies on quasi-triangular channels have been carried out. Vitovsky et al. [11] and Makarov et al. [12] measured wall temperature distributions of He-Xe in the range of 2860 < Re < 44,050, observing abnormal temperature spikes at channel inlets and elevated temperature in the stagnant area near the narrow region. Nakoryakov et al. [13] further demonstrated that the heat transfer capacity of He-Xe in quasi-triangular channels is lower than that in circular tubes due to flow stagnation in narrow regions. Computational Fluid Dynamics (CFD) provides an alternative cost-effective way to gain insights into flow and temperature fields of He-Xe in quasi-triangular channels. Sun et al. [14] revealed significant temperature stratification in narrow gaps, with fluid temperature exceeding the bulk value. Qin et al. [15] conducted simulations with a custom-designed quasi-triangular geometry, and a significant non-uniform circumferential distribution of quasi-triangular pipe wall temperature was observed. Makarov et al. [12] conducted numerical simulations based on the operating condition from the experiment, obtaining temperature distributions that matched the experimental wall temperature results.

Reynolds-averaged Navier–Stokes (RANS) is a widely employed CFD method due to its cost-effectiveness and computational efficiency. RANS employs turbulence models to close the momentum conservation equations and utilizes turbulent Prandtl number (Pr_t) models to establish the relationship between the Reynolds stress and turbulent heat flux and close the energy conservation equation.

Numerous studies have utilized RANS for He-Xe mixture simulations. When employing RANS, identifying accurate and universally applicable turbulence models and Pr_t models remains a critical research objective. Table 1 summarizes the turbulence and Pr_t models adopted in existing studies. Most studies assess turbulence models solely against circular tube experiments, with the majority recommending the SST k-ω model [14–20]. Other modeling approaches, including the Reynolds Stress Model (RSM), Realizable k-ε (RKE), Standard k-ε, and Transition SST, have also been adopted in research [12,21–24]. Regarding Pr_t selection, given the similar low-Pr characteristics exhibited by He-Xe mixtures and liquid metals, researchers usually substitute constant Pr_t with liquid metal-derived Pr_t models [25–27]. Several studies neglect the influence of Pr_t and retain default values, while others modify Pr_t using either liquid metal models or Zhou's model [16].

Table 1. Summary of previous numerical simulations on heat transfer of helium–xenon gas mixture.

Researcher	Turbulence Model	Pr_t Model	Channel
Meng [21]	Realizable k-ε	0.85	Circular
Zhou [16]	SST k-ω	Zhou/Kays	Circular
Qin [15]	SST k-ω	Weigand	Circular
Sun [14]	SST k-ω	/	Quasi-triangular
Makarov [12]	Transition SST	Kays	Quasi-triangular
Ning [22]	BSL-RSM	/	Rod bundle
Wang [23]	Standard k-ε	1.5	Circular/rod bundle
Hou [17]	SST k-ω	0.85	Quasi-triangular
Li [18]	SST k-ω	Zhou	Rod bundle
Wang [19]	SST k-ω	Zhou	Circular
Yang [20]	SST k-ω	Kays	Circular/quasi-triangular
Ning [24]	Realizable k-ε	Kays	Circular/quasi-triangular

Notably, the majority of simulations focus on circular tubes, with quasi-triangular channels receiving limited attention. Simulations in circular channels reveal significant discrepancies in predicted wall temperature when applying different turbulence models and Pr_t models [16]. As the most commonly used turbulence model, SST k-ω cannot predict the heat transfer deterioration at the near-inlet zone of the quasi-triangular channel [18]. The models referenced in existing studies will inform this work. However, most exhibit limitations for quasi-triangular channel applications. Conventional models fail to accurately simulate the distinct flow and heat transfer phenomena in quasi-triangular channels compared to circular tubes. Present Pr_t models were developed for liquid metals; however, their applicability to He-Xe remains uncertain. The Zhou model, based on He-Xe experimental data, has been validated only for circular channels. Current research indicates that comprehensive evaluation of models for He-Xe simulations remains incomplete. Critical analyses for quasi-triangular channels lack rigorous grid independence verification and systematic model evaluation.

This article is organized as follows: Section 2 describes the experimental description and set up of numerical simulations. In Section 3, the predicted performance of turbulence models is compared to identify optimal choices for He-Xe flow simulation. In Section 4, different Pr_t models using optimal turbulence models are tested to reveal low-Pr He-Xe Pr_t distribution patterns and the optimal Pr_t model. The present study is the first systematic evaluation of turbulence models and Prt models for He-Xe gas mixture heat transfer using a quasi-triangular channel with experimental validation, providing a foundation for improving thermal–hydraulic prediction accuracy in tight-lattice rod bundles and for advanced reactor thermal–hydraulic analysis.

2. Methodology

2.1. Numerical Method

The behavior of fluid flow is governed by a set of fundamental conservation laws, namely the conservation of mass, momentum, and energy. These principles form the foundation for analyzing and predicting flow characteristics across various engineering applications. In this investigation, the finite volume method (FVM) is adopted to discretize and solve the Reynolds-averaged Navier–Stokes (RANS) equations under steady-state conditions. The FVM is particularly well suited for this purpose due to its inherent conservation properties and flexibility in handling complex geometries. Within the finite volume method, the conservation laws for mass, momentum, and energy are applied to discretized control volumes, resulting in their integrated form. The Reynolds-averaged Navier–Stokes equations are as follows:

$$\frac{\partial}{\partial x_i}(\rho u_i) = 0, \tag{1}$$

$$\frac{\partial \rho u_i u_j}{\partial x_j} = -\frac{\partial P}{\partial x_i} + \frac{\partial}{\partial x_j}\left[\mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3}\delta_{ij}\frac{\partial u_l}{\partial x_l}\right)\right] + \frac{\partial}{\partial x_j}\left(-\rho\overline{u_i' u_j'}\right), \tag{2}$$

$$\frac{\partial}{\partial x_i}\left[u_j(\rho E + P)\right] = \frac{\partial}{\partial x_i}\left(\lambda\frac{\partial T}{\partial x_i}\right) + \frac{\partial}{\partial x_i}\left(-\rho c_P\overline{u_i' T'}\right). \tag{3}$$

Two additional terms, known as Reynolds stress ($u_i' u_j'$) and turbulent heat flux ($u_i' T'$), require appropriate models for their incorporation. The overbars on the Reynolds stress and turbulent heat flux terms indicate that these terms represent the statistical average of the momentum/heat transport effects caused by the turbulent fluctuations. Within the RANS model, the critical distinction between turbulence closure approaches lies in the treatment of Reynolds stress tensor modeling. The Reynolds Stress Model (RSM) explicitly solves seven additional transport equations to resolve anisotropic turbulence characteristics, albeit at an increase in computational cost compared to eddy viscosity models. This stands in contrast to the Boussinesq hypothesis-based closure, where the Reynolds stress tensor is linearly related to the mean strain rate tensor through the turbulent viscosity (μ_t). According to the Boussinesq hypothesis, the Reynolds stress item can be expressed as follows:

$$-\rho\overline{u_i' u_j'} = \mu_t\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) - \frac{2}{3}\rho k\delta_{ij}. \tag{4}$$

In the analysis of viscous flow, a number of turbulence models are commonly employed, which include SST k-ω, RKE, and Transition SST, and they are adopted in this study. Additionally, the RSM, selected in a previous study, will be included in the model comparison.

2.2. Pr_t Models

Pr_t, a dimensionless parameter characterizing the relative efficiency of momentum transport to heat transport in turbulent flows, is defined as the ratio of turbulent kinematic viscosity (v_t) to turbulent thermal diffusivity (α_t).

When $\mathrm{Pr}_t = 1$, the diffusion efficiencies of turbulent momentum and heat are comparable. For $\mathrm{Pr}_t > 1$, momentum diffusion dominates, resulting in a temperature boundary layer thicker than the velocity boundary layer. In the Reynolds-averaged energy conservation equation, the turbulent heat flux term is closed via the Simple Gradient Diffusion Hypothesis (SGDH):

$$-\rho c_P \overline{u_i' T'} = \rho c_P \alpha_t \frac{\partial T}{\partial x_i}. \quad (5)$$

Here, α_t denotes the turbulent thermal diffusivity, and Pr_t is defined as follows:

$$\mu_t = \rho v_t, \quad (6)$$

$$\mathrm{Pr}_t = \frac{v_t}{\alpha_t}. \quad (7)$$

Consequently, the turbulent heat flux can be expressed as follows:

$$-\rho c_P \overline{u_i' T'} = \frac{c_P \mu_t}{Pr_t} \frac{\partial T}{\partial x_i}. \quad (8)$$

The selection of Pr_t directly influences heat flux predictions. For high-Pr fluids ($\mathrm{Pr} \gg 1$), Pr_t asymptotically approaches unity, and engineering computations typically adopt a constant value (the default $\mathrm{Pr}_t = 0.85$ in ANSYS Fluent 2021). However, for low-Pr fluids, Pr_t exceeds 1. Furthermore, Pr_t increases as Pr decreases. Conventional turbulence models introduce distortions in convective heat transfer predictions for helium–xenon mixtures due to their neglect of this critical low-Pr fluid behavior.

2.3. Physical Properties of He-Xe Gas Mixture

The thermophysical properties of the working fluid are pivotal to numerical simulation accuracy. This study utilizes a He-Xe gas mixture with a molar mass of 40 g/mol as the working fluid, which is used in Makarov's experiment [12]. Distinct from conventional monatomic gases, the He-Xe mixture exhibits non-ideal gas behavior due to its elevated molecular mass and density, rendering the Chapman–Enskog theory under ideal gas assumptions inapplicable for property calculations. Consequently, transport properties were determined using the multiscale intermolecular potential model developed by Tournier et al. [28]. Validated across extended operational regimes (temperature $\leq$ 1400 K, pressure $\leq$ 20 MPa), the thermophysical properties of He-Xe in this study were computed via these correlations. Table 2 details the thermal properties of He-Xe with a molar mass of 40 g/mol at the operating pressure, where C_p is treated as constant due to its negligible temperature dependence.

Table 2. Thermal properties of helium–xenon gas mixture.

Pressure (Pa)	Properties	Value
127,750	ρ (kg/m^3)	$9.45425 - 5.558 \times 10^{-2}T + 1.62363 \times 10^{-4}T^2 - 2.58441 \times 10^{-7}T^3 + 2.27439 \times 10^{-10}T^4 - 1.03762 \times 10^{-13}T^5 + 1.91296 \times 10^{-17}T^6$
	C_p (J/kg·K)	525.8
	λ (W/m·K)	$0.01054 + 2.11549 \times 10^{-4}T - 8.46116 \times 10^{-8}T^2 + 2.24425 \times 10^{-11}T^3$
	μ (kg/m·s)	$-1.04667 \times 10^{-6} + 1.01515 \times 10^{-7}T - 4.83718 \times 10^{-11}T^2 + 1.35794 \times 10^{-14}T^3$

2.4. Experiment Descriptions

Makarov's experiment with a He-Xe mixture in a quasi-triangular channel is the benchmark for validation. Figure 1 presents the geometric model of this study [12]. In Figure 1, the curvature radius of the outer wall of the channel is R = 6 mm, with a wall thickness δ = 0.3 mm. T_{wi}, located in the mid-concave region of the channel, is the data point representing wall temperature measured by thermocouples. The original channel consists of three components: unheated round tube sections at the inlet and outlet, and a quasi-triangular tube section in the middle. The experimental quasi-triangular tube is fabricated from thin-walled nickel–chromium alloy with a radius of 6 mm, approximating a triangular shape. This study employs the same nickel–chromium alloy with identical physical properties as the channel material for the numerical simulations. The hydraulic diameter (D_e) is 1.42 mm, and the length (L) is 450 mm. The experimental cross-section comprises a nickel–chromium alloy tube with a wall thickness of 0.3 mm, which was electrically heated to maintain a constant heat flux density. As the literature lacks explicit geometric parameters for the connection region between circular and quasi-triangular tubes, simulation results cannot accurately capture wall temperature distribution outside the quasi-triangular domain. Consequently, this study only models the heated quasi-triangular pipe section and compares wall temperature in this region.

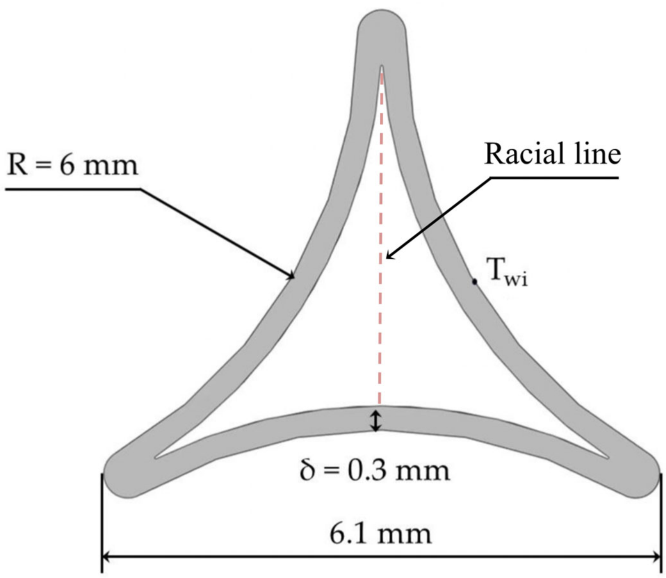

Figure 1. Scheme of experimental pipe [12].

The experimental study presents wall temperature data under four operating conditions. All experimental wall temperature data were obtained from the mid-concave region of the quasi-triangular channel at the T_{wi} point marked in Figure 1. The set of these conditions is presented in Table 3, obtaining boundary conditions of the He-Xe mixture with a molar mass of 40 g/mol in heated and unheated quasi-triangular channels.

Table 3. Boundary conditions of experiment [12].

No.	P_{in} (Pa)	T_{in} (K)	G (kg/s)	ρ_{in} (kg/m^3)	q_w (W/m^2)	Re
Condition 1	127,750	300.5	6×10^{-4}	2.019	1932	5536
Condition 5	311,500	298.9	2.4×10^{-3}	4.952	9845	22,143
Condition 6	403,000	298.7	3.2×10^{-3}	6.410	12,993	29,524
Condition 7	529,600	298.2	4.31×10^{-3}	8.436	17,320	39,717

2.5. Boundary Conditions and Solution Methods

The numerical simulations were predicated on these fundamental assumptions: the working fluid exhibits incompressible behavior within the operational parameters of interest, and body forces including gravitational acceleration are neglected. Previous research confirms significant effects of conjugate heat transfer on thermal characteristics in quasi-triangular channels [17]; the numerical simulations in this study were conducted by employing conjugate heat transfer methodology. A second-order upwind discretization scheme was systematically applied to both convective and diffusive terms in the governing equations to preserve numerical accuracy. Pressure–velocity coupling was resolved through implementation of the SIMPLEC (Semi-Implicit Method for Pressure-Linked Equations) algorithm. Solution convergence was rigorously defined by achieving normalized residuals below 1×10^{-4} for continuity, momentum, and energy equations. To ensure numerical stability during steady-state computations, the area-weighted average static temperature at the outflow boundary was monitored, with temporal fluctuations constrained within $\pm 0.5\%$ of the mean value. This monitoring protocol provided quantitative verification of solution stationarity.

In this study, FLUENT 2021 software is employed for the resolution of the governing equations [29]. The Pr_t models are established via user-defined functions (UDFs) in CFD simulations. The He-Xe gas mixture is selected as the primary coolant. The boundary conditions are defined as follows: the inlet and outlet of the channel are designated as pressure inlet and mass flow outlet. The walls of the quasi-triangular channel are treated as no-slip boundary conditions and are subjected to uniform heating. Experimental and numerical validation reveals that subsonic flow phenomena occurred in all experimental conditions except Condition 1. Therefore, Condition 1 is selected as the validation case for the present study.

2.6. Mesh Sensitivity Study

A refined structured mesh was generated using ICEM, as shown in Figure 2. The mesh is refined near the wall with a first-element height of 0.00013 mm. The dimensionless wall distance (y^+) is always less than 1. The thickness of the first grid layer is consistent with the boundary layer thickness. This structured meshing strategy provides a high-precision computational foundation for subsequent analysis of heat transfer mechanisms in the He-Xe mixture.

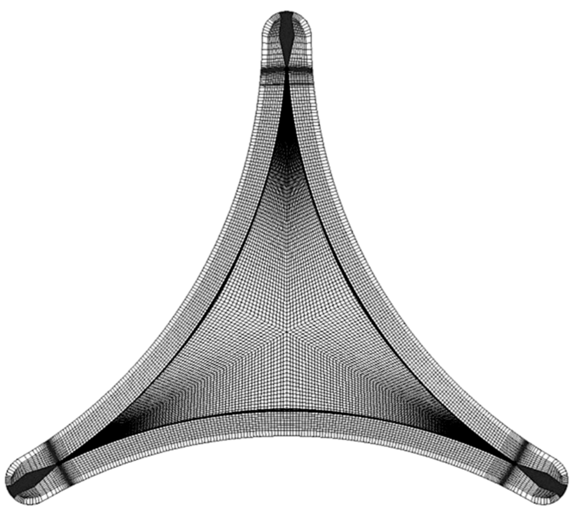

Figure 2. Mesh diagram of simulation model.

To confirm that the grid count does not affect the computational results, this study conducted grid independence verification across four grid configurations (with total cell counts of 0.9 million, 1.8 million, 3.6 million, and 7.2 million, as shown in Figure 3). The grid independence test results demonstrate that the temperature along the characteristic line gradually decreases with increasing grid density. Notably, when the grid count exceeds 3.6 million, coolant temperature distribution becomes independent of grid size. Consequently, a grid configuration with 3.6 million cells was ultimately adopted. These results effectively validate the reliability of the numerical model.

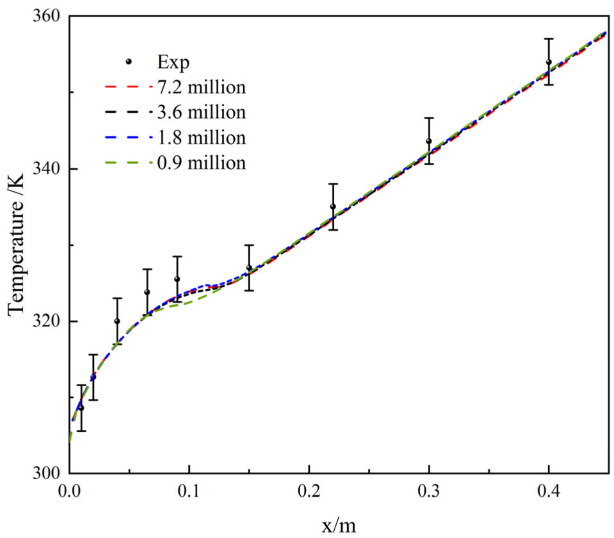

Figure 3. Wall temperature under different grid numbers.

3. Performance of Turbulence Models

3.1. Turbulence Model Study

Based on the experimental wall temperature result, the quasi-triangular channel exhibits a distinct temperature spike in its initial section, where the flow develops as a developing flow (hereafter termed the transition region). This temperature elevation correlates with the growth of the laminar boundary layer, which progressively thickens with increasing distance from the inlet. This thickening progressively destabilizes the flow, initiating the transition to turbulence. Notably, the transition region for He-Xe flow in quasi-triangular channels extends significantly, reaching $80D_e$ under this condition. Following the laminar–turbulent transition, the wall temperature immediately decreases slightly and subsequently increases monotonically (hereafter termed the steady region). In contrast to circular tubes, the narrow gap of the quasi-triangular channel contains a flow stagnation zone. Within this zone, the flow remains laminar, delaying the laminar–turbulent transition. Consequently, the wall temperature evolution in quasi-triangular channels reveals a trend markedly divergent from that observed in circular tube experiments. Simulating such a complex flow field requires a more accurate turbulence model. Besides the commonly used SST k-ω model for He-Xe, other models will be compared. These include RKE, Transition SST, and the RSM.

3.2. Results and Discussion

Based on the experimental wall temperature result at the mid-concave region of the quasi-triangular channel, computational results from four turbulence models under Condition 1 were benchmarked against empirical data across nine axial positions, excluding the tenth measurement point due to outlet-induced thermal distortion.

As schematically depicted in Figure 4, computational results for wall temperature at the mid-concave region of the quasi-triangular channel are presented, corresponding to the experimental data obtained at this location. All models exhibit a consistent wall temperature distribution trend that aligns with the experimental observation in the fully developed flow regime. Systematic underprediction relative to experimental values is evident across all simulations, with Transition SST yielding the highest overall prediction and demonstrating the closest agreement with the empirical data in the laminar transition region near the channel inlet. SST k-ω and the RSM generate comparable temperature profiles that converge with the Transition SST result in the mainstream section, while RKE produces the lowest prediction in both the transition and developed region. These discrepancies are attributed to turbulence model-induced variations in flow field development, which correspondingly alter the axial progression of thermal distribution patterns.

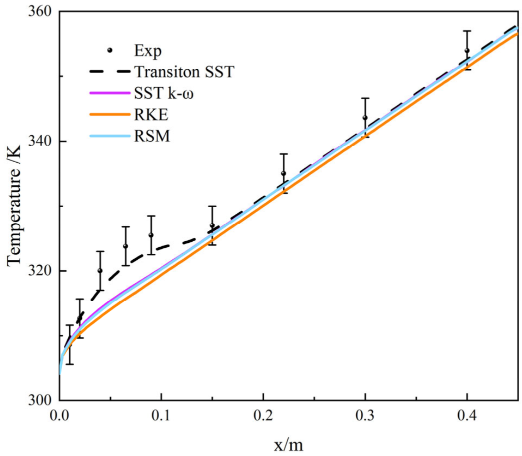

Figure 4. Wall temperature predictions of turbulence model [12].

Table 4 presents the experimental wall temperature values and predicted wall temperature values. Numerical predictions were quantitatively evaluated against experimental measurements using maximum error (ME), mean absolute error (MAE), and root mean square error (RMSE) metrics at corresponding measurement locations. Table 5 presents the error analysis for four turbulence models under Condition 1. The Transition SST model demonstrates superior agreement with the experimental data, exhibiting the lowest values across all error metrics. This ME value is significantly smaller than other models' predictions. The consistently minimal MAE and RMSE values further confirm its exceptional performance. In contrast, the RKE model yields the highest errors in all three metrics. Table 6 details the regional error analysis, confirming that maximum errors consistently occur within the transition region; thus, ME values are excluded from the transition region comparison. The Transition SST model achieves the lowest average error (1.919 K) in the laminar transition zone, significantly outperforming other models. SST k-ω and the RSM exhibit comparable error magnitudes, while RKE demonstrates the poorest performance in this region.

Table 4. Experimental wall temperature values and predicted wall temperature values.

Region	**Transition**						**Steady**		
Data	**Point1**	**Point2**	**Point3**	**Point4**	**Point5**	**Point6**	**Point7**	**Point8**	**Point9**
Experiment	308.63	312.65	320	323.8	325.5	327	335	343.62	354
Simulation									
Transition SST	309.388	312.985	316.912	320.876	323.009	326.226	333.315	341.924	352.542
SST k-ω	308.846	311.389	314.017	317.045	319.401	325.637	333.031	341.672	352.22
RSM	308.742	311.126	313.679	316.759	319.224	325.595	333.053	341.64	352.224
RKE	308.385	310.459	312.758	315.761	318.289	324.712	332.112	340.748	351.32

Table 5. Prediction error of wall temperature for different turbulence models.

Error	Transition SST	SST k-ω	RSM	RKE
ME (K)	3.088	6.755	7.041	8.039
MAE (K)	1.689	3.041	3.153	3.962
RMSE (K)	1.926	3.843	4.001	4.748

Table 6. Prediction error of wall temperature for different turbulence models at different regions.

Region	Error	Transition SST	SST k-ω	RSM	RKE
Transition	MAE (K)	1.919	4.063	4.255	4.985
	RMSE (K)	2.235	4.904	5.123	5.897
Steady	ME (K)	1.696	1.969	1.980	2.889
	MAE(K)	1.403	1.765	1.777	2.682
	RMSE (K)	1.452	1.781	1.791	2.693

In the hydrodynamically stable region, Transition SST, SST k-ω, and the RSM exhibit comparable error magnitudes, with Transition SST demonstrating the closest agreement with the experimental values. The RKE model consistently yields the largest prediction errors. This performance hierarchy substantiates that the SST model—through its k-ω formulation at low Reynolds numbers—provides superior resolution of turbulent characteristics within the boundary layer. Specifically for quasi-triangular channels experiencing laminar boundary layer thickening, the model's enhanced near-wall treatment delivers more accurate flow physics representation.

The Transition SST model further incorporates specialized transition equations that explicitly resolve the laminar–turbulent transition process.

Based on the SST $k - \omega$ model, the Transition SST model introduces two additional transport equations governing the intermittency factor (γ) and the momentum thickness Reynolds number ($\widetilde{Re}_{\theta t}$). These equations enable the model to predict laminar–turbulent transition phenomena.

The transport equation for the intermittency γ is defined as follows:

$$\frac{\partial(\rho\gamma)}{\partial t} + \frac{\partial(\rho u_j \gamma)}{\partial x_j} = P_{\gamma_1} - E_{\gamma_1} + P_{\gamma 2} - E_{\gamma 2} + \frac{\partial}{x_j}\left[\left(\mu + \frac{\mu_t}{\sigma_\gamma}\right)\frac{\partial \gamma}{\partial x_j}\right]. \tag{9}$$

The transport equation for the transition momentum thickness Reynolds number $\widetilde{Re}_{\theta t}$ is as follows:

$$\frac{\partial\left(\rho \widetilde{Re}_{\theta t}\right)}{\partial t} + \frac{\partial\left(\rho u_j \widetilde{Re}_{\theta t}\right)}{\partial x_j} = P_{\gamma_1}\frac{\partial}{x_j}\left[\sigma_{\theta t}(\mu + \mu_t)\frac{\partial \widetilde{Re}_{\theta t}}{\partial x_j}\right]. \tag{10}$$

The transition momentum thickness Reynolds number determines transition onset. The equation predicts transition onset location based on local flow parameters by coupling the boundary layer momentum thickness Reynolds number with the critical Reynolds number (Re_{cr}), thereby accounting for key influencing factors such as pressure gradient and turbulence intensity. The intermittency factor suppresses turbulent kinetic energy production in laminar regions, preventing premature transition triggered by overpredicted turbulent viscosity in the SST k-ω model.

In quasi-triangular channels, thicker laminar boundary layers amplify the impact of premature transition. Other turbulence models lack this suppression mechanism, leading to early overprediction of turbulent viscosity during boundary layer development and causing enhanced turbulent heat transfer and lower wall temperature in simulations.

These factors of Transition SST enable more precise prediction of thermal variations in the upstream transition zone, where other turbulence models exhibit significant shortcomings. Consequently, the Transition SST model emerges as the optimal turbulence model for such geometrically complex configuration. Subsequent research will employ the Transition SST model as the turbulence model for Pr_t model investigations.

4. Performance of Pr_t Models

4.1. Pr_t Model Study

Numerical simulations of circular tube experiments using various turbulence models reveal substantial discrepancies between predicted and experimentally measured wall temperatures across different operating conditions. The constant Pr_t = 0.85 model, derived from conventional fluids (Pr~1), demonstrates inadequate predictive accuracy for low-Pr fluids, necessitating the development of advanced Pr_t models.

The earliest Pr_t models, developed for liquid metals, account for multiple parameters influencing Pr_t and have established a theoretical foundation for modeling low-Pr fluids [25–27]. Existing Pr_t models which have been commonly used in liquid metal studies or preliminarily evaluated in He-Xe studies are summarized in Table 7.

Table 7. Existing Pr_t models.

Pr_t Models	Correlation
Fluent	$\mathrm{Pr}_t = 0.85$
Reynolds analogy	$\mathrm{Pr}_t = 1.0$
Jischa [29]	$\mathrm{Pr}_t = 0.9 + \frac{182.4}{\mathrm{Pr}\cdot\mathrm{Re}^{0.888}}$
Aoki [30]	$\mathrm{Pr}_t{}^{-1} = 0.014\mathrm{Re}^{0.45}\mathrm{Pr}^{0.2}\left[1 - \exp\left(-\frac{1}{0.014\mathrm{Re}^{0.45}\mathrm{Pr}^{0.2}}\right)\right]$
Reynolds [31]	$\mathrm{Pr}_t = \left(1 + 100\mathrm{Pe}^{-0.5}\right)\left(\frac{1}{1+120\mathrm{Re}^{-0.5}} - 0.15\right)$
Kays [25]	$\mathrm{Pr}_t = 1/\left\{\frac{1}{1.7} + \mathrm{CPe_t}\sqrt{\frac{1}{0.85}} - (\mathrm{CPe}_t)^2 \times \left[1 - \exp\left(\frac{-1}{\mathrm{CPe}_t\sqrt{0.85}}\right)\right]\right\}$
Weigand [27]	$\mathrm{Pr}_t = 1/\left\{\frac{1}{2\mathrm{Pr}_{t,\infty}} + \mathrm{CPe_t}\sqrt{\frac{1}{\mathrm{Pr}_{t,\infty}}} - (\mathrm{CPe}_t)^2 \times \left[1 - \exp\left(\frac{-1}{\mathrm{CPe}_t\sqrt{\mathrm{Pr}_{t,\infty}}}\right)\right]\right\}$ $\mathrm{Pr}_{t,\infty} = 0.85 + \frac{100}{\mathrm{Pr}\cdot\mathrm{Re}^{0.888}}$
Zhou [16]	$\mathrm{Pr}_t = 1/\left\{\frac{1}{2\mathrm{Pr}_{t,loc}} + \mathrm{CPe_t}\sqrt{\frac{1}{\mathrm{Pr}_{t,loc}}} - (\mathrm{CPe}_t)^2 \times \left[1 - \exp\left(\frac{-1}{\mathrm{CPe}_t\sqrt{\mathrm{Pr}_{t,loc}}}\right)\right]\right\}$ $\mathrm{Pr}_{\mathrm{t,loc}} = 0.86 + \frac{30}{\mathrm{Pr}\cdot\mathrm{Re}_{loc}^{0.888}}$ $\mathrm{Re}_{loc} = \frac{\rho_{loc}u_{loc}D}{\mu_{loc}}$

Reynolds analogy, based on the assumption that the flow and thermal boundary layer thicknesses are comparable, represents the simplest model for turbulent heat transfer. Its accuracy is optimized by varying the Pr_t number.

Given the similar low-Pr characteristics exhibited by He-Xe mixtures and liquid metals, researchers usually substitute constant Pr_t with liquid metal-derived Pr_t models to enhance prediction accuracy. However, the applicability of extrapolating these models to He-Xe mixtures remains questionable, as the Pr of He-Xe mixtures is one order of magnitude higher than that of liquid metals.

Because of the strong dependence of Pr_t on Pr and Re for low-Pr fluids, efforts have been made in the past to develop two-parameter Pr_t models which are able to account for the dependence of Pr_t on Pr as well as Re [29,30]. Jischa developed a Pr_t model from the modeled transport equations for turbulent kinetic energy and turbulent heat flux. Due to the spatial influence, the model can be seen as a mean value of Pr_t across the whole boundary layer:

$$Pr_t = 0.9 + \frac{182.4}{Pr \cdot Re^{0.888}}. \tag{11}$$

To take the dependence of Pr_t on the wall distance into account, Kays developed a three-parameter Pr_t model [25]:

$$Pr_t = 1/\left\{\frac{1}{1.7} + CPe_t\sqrt{\frac{1}{0.85}} - (CPe_t)^2 \times \left[1 - \exp\left(\frac{-1}{CPe_t\sqrt{0.85}}\right)\right]\right\}. \tag{12}$$

C = 0.3 is a constant determined by liquid metal experimental data, prescribing the spatial distribution of Pr_t and Pe_t. The Kays model has a disadvantage that Pr_t is always smaller than 1.7, unable to adjust the Pr_t of different low-Pr fluids. On the basis of the Kays model, Weigand modified the original Kays model by replacing the constant value 0.85 with $Pr_{t,\infty}$ [27]:

$$Pr_{t,\infty} = 0.85 + \frac{D}{Pr \cdot Re^{0.888}}. \tag{13}$$

$Pr_{t,\infty}$ uses the functional form or Jisacha as an approximation for $Pr_{t,\infty}$. A value for the constant D of 100 is determined by liquid metal experimental data. Further adjusting of the constant D might achieve better agreement between simulations and measurements.

Among these, Zhou developed a Pr_t model specifically for He-Xe mixtures based on the Kays model and Taylor's circular tube experimental data, which has been subsequently adopted in multiple studies. In the Zhou model, locally computed Re is proposed to replace Re [16]:

$$Re_{loc} = \frac{\rho_{loc} u_{loc} D_e}{\mu_{loc}}. \tag{14}$$

Re_{loc} can represent the influence of physical properties and flow state on Pr_t. It is introduced in the $Pr_{t,loc}$ function, where constants are determined by Taylor's experiment data, as shown below:

$$Pr_{t,loc} = 0.86 + \frac{30}{Pr \cdot Re_{loc}^{0.888}}. \tag{15}$$

In the present study, multiple Pr_t models have been proposed, with three models identified as commonly used for numerical simulations of He-Xe mixtures: the Kays model, the Weigand model, and the Zhou model. Other Pr_t models are not applicable for He-Xe simulations [16].

4.2. Results and Discussion

Figure 5 displays the wall temperature predictions, with the Reynolds analogy, Kays model, Weigand model, and Zhou model using the Transition SST model for quasi-triangular channel simulations.

Application of the Reynolds analogy revealed that setting Pr_t to Pr_t = 1.2 minimizes errors in fully developed flow region prediction compared to experimental data, while also reducing discrepancies in the transition region. Compared to the default Pr_t = 0.85, this adjustment yields significant accuracy improvements

The Kays model, derived analytically as a function of Pr and fitted into a semi-empirical three-parameter model, better characterizes the variation in Pr_t with Pr, Re, and dimensionless wall distance (y^+) for He-Xe mixtures.

From the wall temperature comparison curve, the Kays model demonstrates superior performance over simulations without Pr_t models in both the transition region and fully developed flow regions. Its structural formulation captures Pr_t dependencies on Re and y+ more effectively under low-Pr He-Xe conditions.

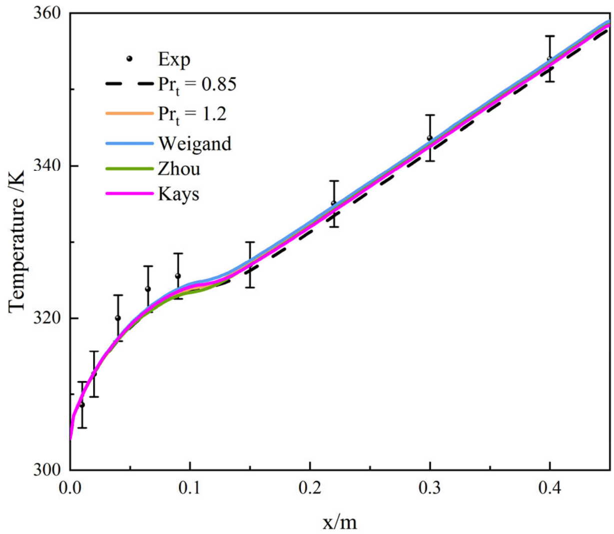

Figure 5. Wall temperature predictions of Pr_t models.

In the transition region, the model shows marginal improvement over baseline predictions, while achieving significant optimization in the fully developed regime. The Pr_t of He-Xe mixture exceeds 0.85 in the near-wall region, and using the default value amplifies the turbulent thermal diffusivity effects, underestimating temperature predictions.

Comparing the Kays model with Pr_t = 1.2, the former yields predictions closer to the experimental data in the transition region, while both produce comparable results in fully developed zones. The Kays model predicts slightly lower temperatures in the fully developed region, where Pr_t = 1.2 aligns more closely with true values. This indicates that the Re and y+-dependent formulation in the Kays model enhances accuracy in high-velocity-gradient transition zones, whereas parameter constraints and constant settings lead to underestimation in fully developed regions.

Selecting two transition zones (x = 0.065, x = 0.09) and two fully developed zones (x = 0.15, x = 0.3), radial Pr_t distributions along the concave-to-narrowest-section midline are shown in Figure 6 (left to right: channel concave region to top). In Figure 6, x represents the axial distance from the channel inlet. The mainstream Pr_t from the Kays model ranges between 0.85 and 1.2, increasing to 1.7 near walls, bounded by the model's maximum Pr_t limit. In lower channel sections, the slope of Pr_t variation from wall to mainstream steepens as flow transitions to a fully developed regime, reflecting the thinning and stabilization of boundary layers, while mainstream Pr_t gradually decreases. In upper narrow regions, Pr_t approaches 1.7 and remains nearly constant due to minimal flow velocity, indicating invariant boundary layer thickness in these zones.

Building on the Kays model, the Weigand model introduces the concept of "far-wall Pr_t", eliminating the fixed upper limit of Pr_t < 1.7 inherent in the Kays model. This modification allows for greater flexibility in adjusting numerical values for turbulent core regions. From the wall temperature comparison curves, the Weigand model demonstrates significantly superior performance over simulations without Pr_t models in both the transition region and fully developed flow regions.

Compared with predictions using Pr_t = 1.2, the Weigand model achieves closer alignment with experimental data in the transition region, showing marked improvement. In the fully developed regime, the results are moderately optimized, further approaching experimental measurements. These findings confirm that the Weigand model's refinement of the Kays formulation substantially enhances predictive accuracy for helium–xenon mixtures. It better resolves Pr_t values in both turbulent core and near-wall regions, thereby improving turbulent heat transfer prediction capabilities.

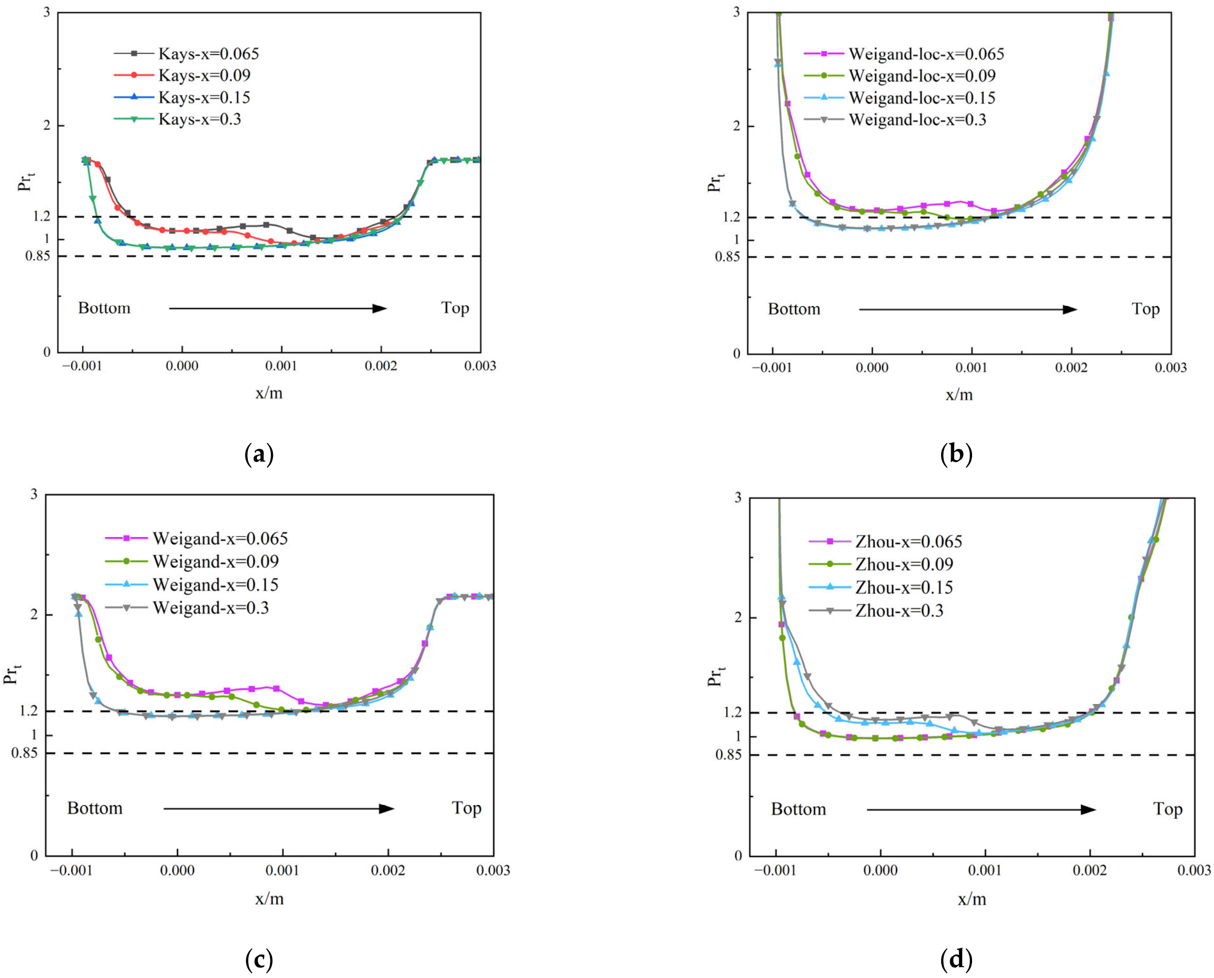

Figure 6. Racial Pr_t in different zones with different Pr_t models. (**a**) Racial Pr_t in different zones with Kays model; (**b**) racial Pr_t in different zones with Weigand model; (**c**) racial Pr_t in different zones with local Re-calculated Weigand model; (**d**) racial Pr_t in different zones with Zhou model.

However, notable discrepancies persist between predicted and experimental temperatures in the transition region. This is attributed to flow stagnation phenomena in narrow channel sections, where increased boundary layer thickness and complex heat transfer behavior challenge the model's predictive capacity.

Existing research indicates that channel geometry causes significant differences in cross-sectional velocity [12]. Fixed Reynolds numbers are replaced with locally computed instantaneous Reynolds numbers for comparison. As is shown in Figure 7, computational errors decrease when using local Reynolds numbers.

The comparison shows that using locally computed Re instead of fixed Re gives a similar prediction result. However, near the end of the transition region, it provides a better wall temperature trend. In quasi-triangular channels, low-Re zones in narrow areas occupy more space radially than in circular tubes. Using locally computed Re allows for flexible handling of these zones and captures changes in Pr_t values there.

The radial Pr_t distribution from the Weigand model using local Re (Figure 6b) shows that mainstream Pr_t centers around 1.2, gradually decreasing as flow stabilizes axially. Near walls, Pr_t growth accelerates due to the absence of an upper constraint in the model formulation. With vanishing turbulent viscosity near walls ($Pe_t \sim 0$), Pr_t rapidly escalates.

As axial flow develops, the lower and upper channel sections exhibit trends similar to the Kays model, differing only in the lack of an explicit Pr_t ceiling.

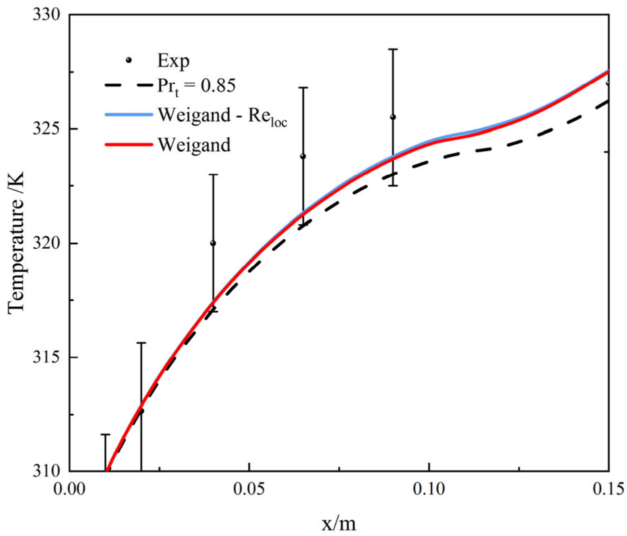

Figure 7. Comparison of local Re-calculated Weigand model with Weigand model.

As is shown in Figure 6c, comparing the fixed Re and local Re Pr_t results reveals that fixed Re simulations produce systematically higher mainstream Pr_t, while constrained Re suppresses near-wall escalation. However, this adjustment shows minimal impact on wall temperature predictions.

The Zhou model incorporates local Re as a parameter to calculate localized Pr_t, replacing the wall-distant Pr_t formulation in the Weigand model. By modifying constants through calibration against circular tube experimental data, the model refines near-wall Pr_t via local Re corrections, thereby more comprehensively accounting for Pr_t variations induced by flow state parameters and reflecting the abrupt near-wall Pr_t escalation observed experimentally. Finally, the model refines far-wall Pr_t parameters based on Taylor's experimental dataset to establish the complete model.

From the predictive results, the modified Zhou model exhibits Pr_t underestimation, particularly evident in the transition region. In this region, its predictions align closely with the constant Pr_t = 0.85 case, producing premature transition point predictions compared to other models, while delivering improved performance over Pr_t = 0.85 in fully developed flow regions. When compared to Pr_t = 1.2 predictions, both transition and fully developed flow region results from the Zhou model show systematically lower Pr_t values, directly contributing to its reduced magnitude.

Figure 6d presents radial Pr_t distributions from Zhou model simulations. The mainstream Pr_t values range between 0.85 and 1.2, consistent with the Kays model results. Near walls, Pr_t rapidly increases due to the model's removal of the maximum Pr_t constraint, allowing for unbounded near-wall growth. Notably, the Pr_t distribution trends oppose those of the Kays model: in the lower channel region, transition-to-fully developed flow evolution shows decreasing wall-to-mainstream Pr_t gradient slope alongside increasing mainstream Pr_t values, whereas the upper narrow region exhibits rapid Pr_t escalation in both flow regimes, ultimately plateauing at ~1.7 due to minimal flow velocity. This stagnation suggests invariant boundary layer thickness in constricted areas.

These Pr_t distributions explain the wall temperature predictions: in the transition region, reduced Pr_t values cause underpredicted temperatures compared to Pr_t = 1.2 (approaching Pr_t = 0.85 result), while the subsequent Pr_t recovery in mainstream regions improves accuracy relative to the constant Pr_t = 0.85 framework.

Table 8 presents the error analysis comparing wall temperature predictions when the Transition SST model is implemented with distinct Pr_t models: the Reynolds analogy, Kays, Weigand, and Zhou models in a quasi-triangular channel. It can be observed that simulations without Pr_t models exhibit the largest error, while incorporating low-Pr fluid-specific Pr_t models significantly improves prediction accuracy compared to the conventional Pr_t = 0.85 assumption. Furthermore, the radial Pr_t distribution stabilizes after transitioning into the fully developed flow region. Error analysis demonstrates that the Weigand model delivers superior temperature prediction accuracy in both transition and steady regions, establishing it as the optimal Pr_t model.

Table 8. Prediction error of wall temperature for different Pr_t models.

Error	Pr_t = 1.2	Kays	Zhou	Weigand
ME (K)	3.016	2.865	2.997	2.772
MAE (K)	1.302	1.318	1.369	1.117
RMSE (K)	1.657	1.595	1.739	1.416

Table 9 shows the error analysis in different regions for the evaluated models, further revealing Weigand model's superior performance across both transition and stable flow regions, delivering the lowest ME, MAE, and RMSE values. The Kays model ranks second in transition regions, with Pr_t = 1.2 in stable zones, a phenomenon attributable to measured $Pr_t \approx 1.2$ in mainstream flow. Conversely, the Zhou model yields the highest errors due to systematic Pr_t underprediction. The Weigand formulation's incorporation of local Re enables marginal accuracy gains in transition regions through flexible low-Re Pr_t distributions, suggesting potential for parametric refinement to optimize transition zone predictions.

Table 9. Prediction error of wall temperature for different Pr_t models in different regions.

Region	Error	Pr_t = 1.2	Kays	Zhou	Weigand
Transition	MAE (K)	1.856	1.755	1.939	1.648
	RMSE (K)	2.142	1.992	2.233	1.855
Steady	ME (K)	0.838	1.072	0.930	0.565
	MAE (K)	0.609	0.772	0.658	0.454
	RMSE (K)	0.662	0.873	0.756	0.463

For quasi-triangular channels, the Transition SST model outperforms alternative turbulence closures by accurately resolving transition phenomena, significantly enhancing flow field predictions. When coupled with the optimized Pr_t model (e.g., Weigand), this model elevates thermal field prediction fidelity while maintaining computational robustness.

5. Conclusions

This study systematically evaluates the performance of multiple turbulence models and Pr_t models for He-Xe mixture flow and heat transfer in quasi-triangular channels using experimental data as the benchmark. This is the first systematic evaluation of both turbulence models and Pr_t models for He-Xe in quasi-triangular channels with experimental validation.

A comparison of turbulence models demonstrates that the Transition SST model outperforms other turbulence models in prediction. It captures the transitional flow phenomena inherent to a quasi-triangular channel, where its adaptability becomes a critical factor for accurate temperature distribution prediction, enabling superior wall temperature predictions compared to alternative turbulence models.

Among Pr_t models, the Weigand model achieves the smallest deviation from experimental wall temperature measurements. Compared to unmodified prediction, errors in both the transition region and fully developed flow region are significantly reduced. This indicates that incorporating Pr_t models for turbulent heat transfer in low-Pr He-Xe mixtures improves predictive accuracy, with the Weigand model emerging as the optimal choice for He-Xe mixture numerical simulations. The coupling of the Transition SST model with the Weigand model reduces the MAE in the transition region by 59.4% and decreases the MAE in the steady region by 74.2%, compared to the conventional SST k-ω model.

This study concludes that coupling the Transition SST model with the Weigand model enables numerical simulation of dense rod-bundled channels. This combination captures unique flow characteristics of such geometries while maintaining high predictive precision, demonstrating strong potential for engineering applications. The heat transfer deterioration observed in the transition region, as revealed by the simulation results, has significant implications for reactor design. Deterioration is a major concern in reactor safety analysis. The use of conventional models may overlook this phenomenon, thereby introducing potential safety risks.

Specifically for dense rod-bundled channels sharing geometric similarities with quasi-triangular channels, this validated model pairing delivers enhanced simulation fidelity for He-Xe flow and heat transfer. It provides an essential reference for subsequent thermal–hydraulic design calculations in He-Xe cooled nuclear reactor systems. For reactor coolant channels, accurately predicting near-inlet heat transfer deterioration is imperative for safety. Higher precision numerical simulations directly support optimized and safer nuclear core design. Further research should concentrate on optimizing the Pr_t model. This may be achieved by adjusting model constants using comprehensive He-Xe experimental data, refining the mathematical formulation of the model, and incorporating locally computed Re to enhance flexibility in low-Re regions for improved Pr_t distribution.

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Nomenclature

C_p	specific heat in constant pressure (J/kg·K)
D_e	hydraulic diameter of tube (m)
E	total energy
G	mass flow rate (kg/s)
k	turbulent kinetic energy (m^2/s)
P	operation pressure (Pa)
Pe	peclet number
Pe_t	turbulent peclet number

Pr	Prandtl number
Pr_t	turbulent Prandtl number
q	heat flux (W/m^2)
Re	Reynolds number
Re_{cr}	the critical Reynolds number
$\tilde{Re}_{\theta t}$	the momentum thickness Reynolds number
T	temperature (K)
u	velocity (m/s)
x	axial distance from inlet (m)
y^+	non-dimensional wall distance
Greek letters	
θ	fluctuating temperature (K)
ρ	density (kg/m^3)
μ	dynamic viscosity (Pa·s)
μ_t	turbulent dynamic viscosity (Pa·s)
υ	kinematic viscosity (m^2/s)
υ_t	eddy diffusivity of momentum (m^2/s)
α_t	eddy diffusivity of heat (m^2/s)
λ	thermal conductivity (W/m·K)
δ_{ij}	kronecker function
γ	intermittency factor
Subscripts	
i	axial direction
j	radial direction
t	turbulent
w	wall
∞	turbulent core region
in	inlet
local	local value

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